

Peter A. Lilley

Software Engineer - Local LLM Inference, Language-to-Geometry Systems, Production Tooling

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SUMMARY

I am a software engineer. I build production systems that other professionals use daily. I built a language-to-geometry pipeline. It uses a plain-language interface to generate 3D models and set up engineering deliverables. I use locally hosted LLM inference to handle intent. I use deterministic C# and .NET code to construct and validate geometry. I also build OCR ingestion and automation tools. These tools cut repetitive drafting from 8 to 12 hours to about 30 seconds per cycle. I work in C#, Python, C++17, TypeScript, and Dart.

SKILLS

Languages: C#, Python, C++, Dart, TypeScript, JavaScript, SQL, LISP, VBA

AI and local models: Ollama, LM Studio, Open WebUI, Google APIs, OCR, prompt engineering, text matching, containerizing inference models, cross-platform edge deployments

Databases and Data Systems: SQLite, Firebase, Postgres, XData schemas, data modeling, JSON structuring, graph calculators

Engineering and math: Matrix math, 3D graphics, Express, React, Git, CMake, Linux, macOS Architecture, Docker, Bash scripting for data pipelines

Domain systems and quality: CAD plugins, Revit automation, 3D metadata, ARCore, part dimensioning, coordinate measuring machines, ISO 9001

Networking: Networking Fundamentals, IPv4/IPv6, Subnetting, Cisco Routing & Switching

EXPERIENCE

David Mason & Associates - Staff Technician (CAD automation / Civil 3D tools)

Jul 2024 - Present

St. Louis, MO

- I build CAD generation tools that turn natural-language input into 3D models and drawing setups using local LLMs and C#.
- I built OCR ingestion software to parse, check, and populate existing PDF plan sets, replacing a manual mark-up loop.
- I automate repetitive drafting with Python and Dynamo, cutting cycles from 8 hours to 30 seconds and reducing errors by 25%.
- I design typed metadata schemas that attach to model entities to coordinate standards across disciplines.
- I built C# plugins for Civil 3D to automate layout, audit drawings, and batch-rename title blocks, preventing long loading delays.

Component Bar Products - Quality Engineer - Manufacturing / Precision Machining

Apr 2023 - Jun 2024

O'Fallon, MO

- I programmed and operated coordinate measuring machines to verify precision-machined parts against GD&T specifications.
- I validated dimensional tolerances against CAD models and prints to catch geometry mismatches.
- I enforced ISO 9001 across production lines and managed calibration and traceability of metrology assets.

Heideman & Associates, Inc. - Revit/CAD Technician - Mechanical/Plumbing

May 2018 - Apr 2019

Fenton, MO

- I produced MEP construction documents and built as-built models in Revit and AutoCAD.
- I built automated data libraries for building-system objects and standardized outdated details.

PROJECTS

Digital Twin Pro

Personal product, Free Parameter LLC | Flutter/Dart 3, SQLite, Firebase, Google Gemini API, ARCore | github.com/freeParameterized/digital-twin-pro

- I designed and shipped a cross-platform 3D inventory application to a Google Play beta.
- I architected a custom 3D renderer with a Z-sorted draw queue and custom projection instead of using a game engine.
- I added photo-based item detection via a paid Gemini API.
- I used SQLite to store inventory data locally with a move audit log, QR booklets, and CSV/JSON export.

CAD integration bridge

C++17, Dear ImGui, Windows COM | [local repository](#)

- I built a C++17 Dear ImGui host. It uses COM to bridge BricsCAD, AutoCAD, and Civil 3D. This personal project proved the concept for my later commercial deployments.
- It discovers installed SDKs at runtime, hot-loads LISP, C#, and BRX plugins, and contains about 14,000 lines of code.

Interactive Portfolio (this site)

React, TypeScript, Three.js, Express, Ollama, Docker

- I built a React Three Fiber portfolio over an Express API.
- It retrieves data from a curated corpus and answers questions through a local model, degrading gracefully when offline.
- I containerize inference models and deploy them to Linux, macOS, and Windows edge devices.

EDUCATION

Calculus II (CLEP Credit) - Saint Louis University (SLU)

2018

St. Louis, MO

Accepted by Ranken Technical College for their calculus curriculum.

Coursework toward Latin American Studies with Technical Applications - Missouri University of Science and Technology

2019 - 2020

Rolla, MO

I completed coursework including rigorous, no-calculator College Algebra. It covered determinants, Cramer's rule, and introductory linear algebra concepts. I have technical fluency in Spanish.

A.A.S., Building Systems Engineering Technology - Ranken Technical College

2019

Wentzville, MO